

RAMCO INSTITUTE OF TECHNOLOGY
Department of Electronics and Communication Engineering
Academic Year: 2019- 2020 (Odd Semester)

Innovative Practices Description

UNIT I: INTRODUCTION

Degree, Semester & Branch: V Semester B.E. CSE, EEE, MECH A & B

Course Code & Title: OAN551 Sensors and Transducers

Name of the Faculty member: Mr.S.Vijayakumar

Name of the Topic: Sensor calibration techniques

Name of the Innovative Practice: Mind map

Date & Time: 16.07.2019 & 15 Minutes

ICT Tool Used: Black Board

Description:

A mind map is an easy way to brainstorm thoughts organically without worrying about order and structure. It allows the student to visually structure the ideas which is used to help with analysis and recall.

A mind map is a diagram for representing tasks, words, concepts, or items linked to and arranged around a subject using a non-linear graphical layout that allows the student to build an intuitive framework around a central concept.

Goals (Learning Outcomes):

- The students will be able to identify and visually record current understandings.
- The students will be able to summarize key information, clarify relationships or associations between information and ideas and draw conclusions.

Use of appropriate methods:

Justification for choosing the topic using mind map activity:

A sensor or instrument that is known to be accurate. It can be used to make reference readings for comparison. Most laboratories will have instruments that have been calibrated against NIST standards. These will have documentation including the specific reference against which they were calibrated, as well as any correction factors that need to be applied to the output. Mind map activity helps the students to easily grasp the overview of sensor calibration techniques with the help of graphical representation.

Effective presentation

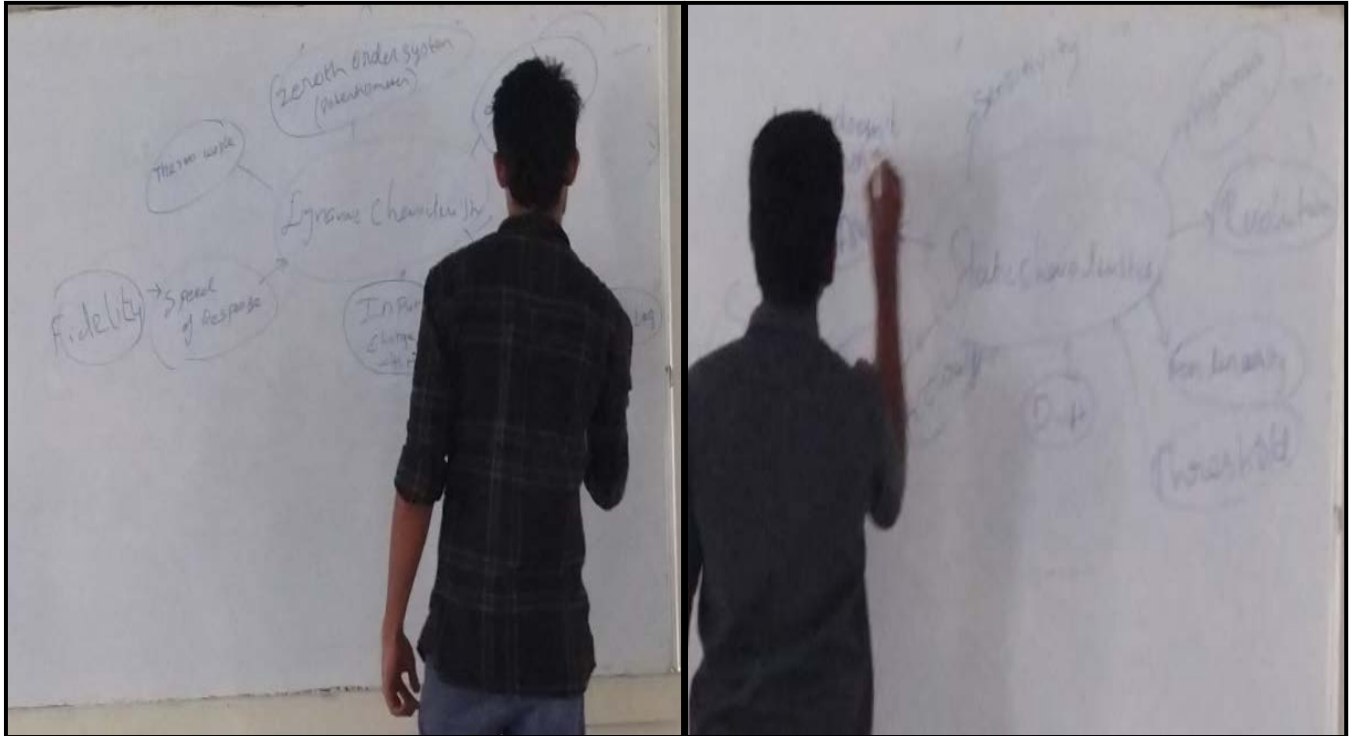
Details of Implementation:

I explained to students about the strategy and ensure that students understand that mind maps are personal representations and as such they are not 'right' or 'wrong'. I Select topic sensor calibration techniques and write this in the centre of a blackboard. Students

then identify connected key words or phrases related to sensor calibration techniques and write these around the topic, progressively moving to less directly related words.

Innovative Teaching Method Execution

Sensor calibration techniques- Mind maps



Significance of Results:

Assessment of Effectiveness/Success of the Activity:

- Every students got equal opportunity to come forward to take part in this activity
- The success of the activity was evaluated by asking the same question in Internal Assessment test II – Around 80% of students answered correctly.

Reflective Critique:

Benefits:

Mind mapping increases creativity and productivity because it's an excellent tool to let students generate more ideas, identify relationships among the different data and information, and effectively improve student's memory and retention. Making a mind map is an excellent way for students to be able to sort through your thoughts and ideas. This activity allows the students to quickly generate creative and even unique ideas in less time. With the help of this activity the students can able to easily differentiate between different calibrations techniques.

Challenges:

I have planned this activity for 10 minutes but few students not able to recall every Details on this topic. So, I gave them another 10 minutes to prepare.

Changes required for future (if required):Nil

Course Outcome	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2	1	1	1	-	-	-	1	-	-	-	1

CO1: The students will be able to expertise in various calibration techniques and signal types for sensors.

References:

1. <https://www.ayoa.com/imindmap/articles/mind-mapping-for-innovation/>
2. D Ernest O Doebelin, "Measurement Systems – Applications and Design", Tata McGraw- Hill, 2009

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Innovative Practices Description

UNIT II: MOTION, PROXIMITY AND RANGING SENSORS

Degree, Semester& Branch: V Semester B.E. CSE, EEE, MECH A & B

Course Code & Title: OAN551 Sensors and Transducers

Name of the Faculty member: Mr.S.Vijayakumar

Name of the Topic: Motion Sensors

Name of the Innovative Practice: Live demonstration

Date & Time: 18.07.19 & 20 Minutes

ICT Tool Used: -

Description:

A method demonstration is a teaching method used to communicate an idea with the aid Of visuals such as flip charts, posters, power point, etc. A demonstration is the process of Teaching someone how to make or do something in a step-by-step process. As you show how, you “tell” what you are doing. A demonstration always has a finished product. The key to a Good demonstration is for the audience to be able to go home and do what you have taught them how to do

Goals (Learning Outcomes):

- The students will be able to identify and visually record current understandings.
- The students will be able to explain about motion sensor.

Use of appropriate methods:

Justification for choosing the topic using mind map activity:

A motion detector is a device that detects moving objects, particularly people. Such a device is often integrated as a component of a system that automatically performs a task or alerts a user of motion in an area. They form a vital component of security, automated lighting control, home control, energy efficiency and other useful systems

Effective presentation

Details of Implementation:

I explained to students about the automatic on-off light using motion sensor with the help of BJT switch. I used LED, a BJT switch, motion sensor and power supply. When the motion sensor detects objects the LED light will be glowing. I used breadboard to make the connection.

Innovative Teaching Method Execution

Motion Sensors- Live demonstration



Significance of Results:

Assessment of Effectiveness/Success of the Activity:

- Every students got equal opportunity to come forward to take part in this activity
- The success of the activity was evaluated by asking the same question in Internal Assessment test II – Around 80% of students answered correctly.

Reflective Critique:

Benefits:

Infusing real world experiences into the instructions can make lectures fresh and enrich learning. Relating and demonstrating through real life situations will make the material easy to understand. It will also draw interest of many students which is the key motivation.

Challenges:

I have planned this activity for 15 minutes but few students did not understand how the BJT acts as a switch. I explained to students about the operation of BJT and how it acts as a switch.

Changes required for future (if required): Nil

Course Outcome	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO2	2	1	1	1	-	-	-	1	-	-	-	1

CO2: The students will be able to apply the various sensors in the Automotive and Mechatronics applications

References:

1. D Ernest O Doebelin, "Measurement Systems – Applications and Design", Tata McGraw- Hill, 2009

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INNOVATIVE TEACHING METHOD

UNIT III: FORCE, MAGNETIC AND HEADING SENSORS

Degree, Semester & Branch: V Semester B.E. CSE, EEE, MECH A & B

Course Code & Title: OAN551 Sensors and Transducers

Name of the Faculty member: Mr.S.Vijayakumar

Name of the Topic: Hall Effect

Name of the Innovative Practice: One minute paper

Date & Duration: 17.08.2019 & 1 minute

ICT Tool Used:- white paper

Description:

A “one-minute paper” may be defined as a very short, in-class writing activity (taking one-minute or less to complete) in response to an instructor-posed question, which prompts students to reflect on the day’s lesson and provides the instructor with useful feedback.

Goals (Learning Outcomes):

- Students will be able to reflect on their understandings and attitudes after completing the Learning experiences from a unit or focus area.
- Students will be able to explain the operation of Hall Effect.

Use of appropriate methods:

Justification for choosing the topic using One minute paper activity:

Hall Effect Sensors are devices which are activated by an external magnetic field. We know that a magnetic field has two important characteristics flux density, (B) and polarity (North and South Poles). The output signal from a Hall Effect sensor is the function of magnetic field density around the device. When the magnetic flux density around the sensor exceeds a certain pre-set threshold, the sensor detects it and generates an output voltage called the Hall Voltage, V_H . I choose this activity, whether my concept and what the students perceived as the main concept are the same. So, I asked to students to write about the Hall Effect after teaching in a piece of paper.

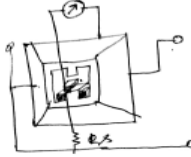
Effective presentation

Details of Implementation:

One minute papers is used at the end of a class to have students reflect back and think more deeply about the most important concept discussed in class that day. I asked the students must first organize their thinking to rank the major points and then decide upon a significant question. Students write a one-minute paper in response to my question. After one minute duration, I collected the papers from students.

Innovative Teaching Method Execution

Hall Effect - One minute paper

953617114063 MECH-B'  Hall effect multiplier $V_H = R_H I B$ $R = \rho l / A = \mu \Omega \text{ m}^{-2}$ $R \propto P$ and $R \propto (V/R_H) dV$ The output voltage of the Hall effect multiplier is proportional to instantaneous power. <u>Adv</u> 1) They can be used for multiple sensor functions like position sensing, speed sensing as well as for sensing the direction of movement too. <u>Dis Adv</u> 1) Accuracy of the measured value is always a concern as external magnetic fields may affect the value. 2) High Temp effect conductor resistance.	M. Apama III CSE Hall effect $E_H = R_H I B$ R_H - Hall effect coefficient * Output voltage is proportional to current and field strength Application :- - Magnetic to electric sensor - Measurement of displacement - Measurement of current - Measurement of power Advantage :- - No wear & tear Dis advantage :- - High temperature affects conductor resistance.
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Significance of Results:

Assessment of Effectiveness/Success of the Activity:

- After conducting this activity I found that many students understand this topic.
- The success of the activity was evaluated by asking the same question in Internal Assessment test II – Around 80% of students answered correctly.

Reflective Critique:

Benefits:

Minute papers are an effective way of involving all students in class simultaneously. It ensures equal participation of each and every class member, including anyone who may be too shy or fearful to participate orally. It can promote class attendance and attentiveness. It can improve the quality of class discussion by having students write briefly about a concept or issue before they begin discussing it.

Challenges:

After teaching this particular topic students asked some time to prepare. So, I gave few minutes to prepare which takes more time to implement this activity than planned.

Changes required for future (if required): Nil

Course Outcome	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO3	1	1	-	-	-	-	-	1	1	1	-	-

CO3: The students will be able to explain force, magnetic and heading sensor in Aerospace and Home appliances

References:

1. <https://oncourseworkshop.com/self-awareness/one-minute-paper/>
2. <https://www.humber.ca/centreforteachingandlearning/instructional-strategies/teaching-methods/classroom-management-designing-the-environment/feedback-in-the-classroom/one-minute-papers.html>
3. Sawney A K and Puneet Sawney, "A Course in Mechanical Measurements and Instrumentation and Control", 12th edition, Dhanpat Rai & Co, New Delhi, 2013.

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Innovative Practices Description

UNIT IV: OPTICAL, PRESSURE AND TEMPERATURE SENSORS

Degree, Semester & Branch: V Semester B.E. CSE, EEE and MECH A & B

Course Code & Title: OAN551 Sensors and Transducers

Name of the Faculty member: Mr.S.Vijayakumar

Name of the Topic: MEMS & Nano Sensors

Name of the Innovative Practice: Animation Video

Date & Duration: 17.09.2019 & 10 minutes

ICT Tool Used: Desktop, projector

Description:

Animation is a brilliant and innovative new way to encourage students to communicate stories, ideas and concepts in a creative and original way. It can be particularly useful as a tool to encourage the creativity of students. Animated videos are a one stop solution that can grab the attention of the students. With its entertaining visuals, it can clearly comprehend the concept. Since it is an analogy to cartoons, students will get hooked to it. By using animation videos, students learn analytical thinking as to how to present a scenario.

Goals (Learning Outcomes):

- The students will be able to easily identify the difference between MEMS and NANO sensors.
- The students will be able to explain the operation of MEMS and NANO sensors.

Use of appropriate methods:

Justification for choosing the topic using animation video activity:

MEMS and Nano sensor. It plays an important role in automotive industry. The student come to know that how silicon is used as base material for MEMS and Nano sensor.

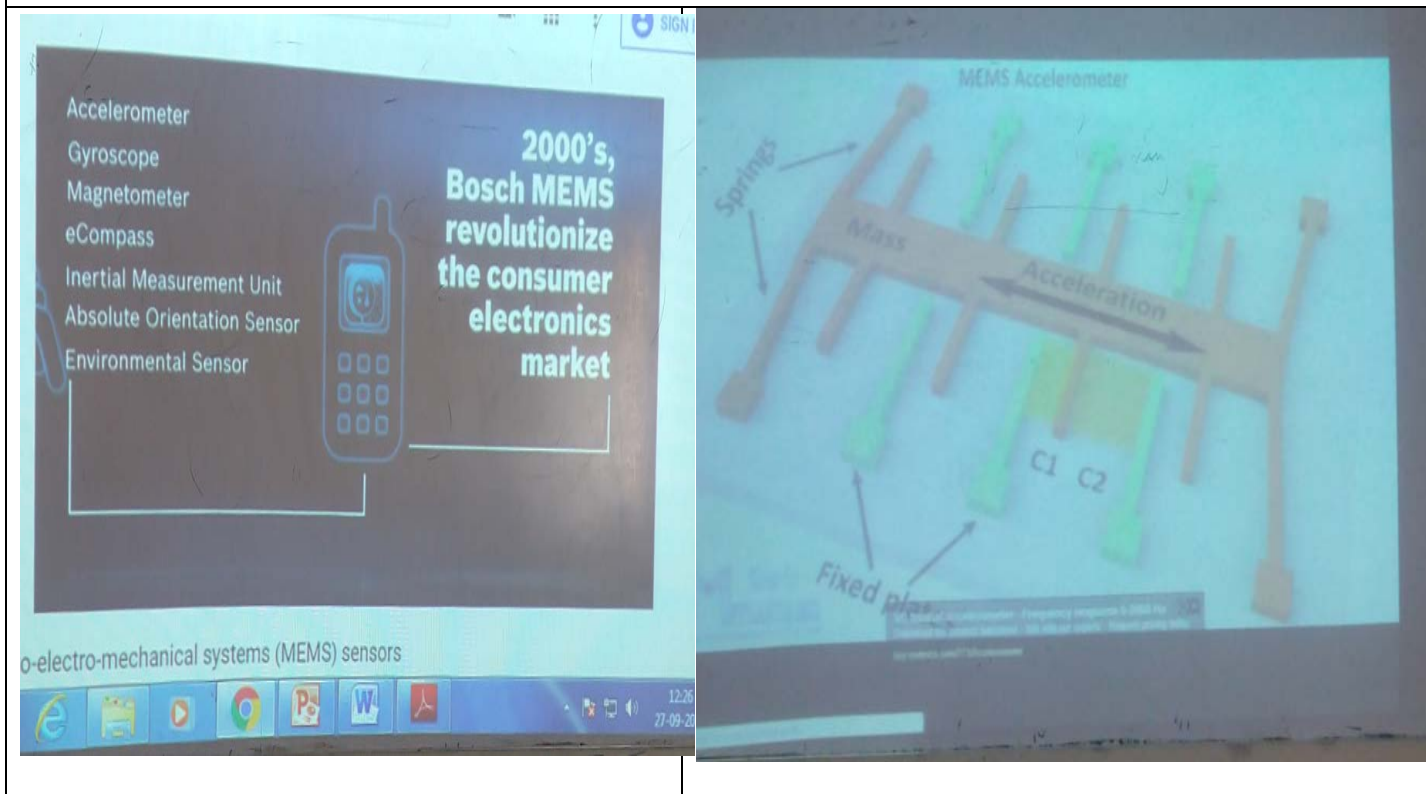
Effective presentation

Details of Implementation:

The suitable animated video for MEMS and NANO sensors was displayed to the students and explained about the different types of MEMS and NANO sensors available in automobile industry.

Innovative Teaching Method Execution

MEMS & Nano Sensors - Animated video



Significance of Results:

Assessment of Effectiveness/Success of the Activity:

- The success of the activity was evaluated by asking the same question in Internal Assessment test III – Around 75% of students answered correctly.

Reflective Critique:

- Benefits:**

The students show more interest by listening to animated video, apart from theory concept. Animation videos help the students to understand the importance of MEMS and NANO sensor.

- Challenges:**

It takes some time to choose relevant videos for the relevant topic in order to make understandable to students.

Changes required for future (if required): Nil

Course Outcome	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO4	2	2	-	-	-	-	-	1	-	-	-	1

CO4: The students will be able to describe the basic principles of various smart sensors

References:

1. Sawney A K and Puneet Sawney, "A Course in Mechanical Measurements and Instrumentation and Control", 12th edition, Dhanpat Rai & Co, New Delhi, 2013.
2. <https://www.youtube.com/watch?v=eqZgxR6eRjo>

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Innovative Practices Description

UNIT V: SIGNAL CONDITIONING and DAQ SYSTEMS

Degree, Semester & Branch: V Semester B.E. CSE, EEE and MECH A & B

Course Code & Title: OAN551 & Sensors and Transducers

Name of the Faculty member: Mr.S.Vijayakumar

Name of the Topic: Applications- Home appliances

Name of the Innovative Practice: Brainstorming

Date & Time: 01.10.19 & 30 Minutes

ICT Tool Used: -

Description:

Brainstorming is a group creativity technique by which efforts are made to find a conclusion for a specific problem by gathering a list of ideas spontaneously contributed by its members.

All the ideas are noted down without criticism and after the brainstorming session the ideas are evaluated.

Goals (Learning Outcomes):

- The students will be able to develop creative thinking processes and problem solving skills.
- The students will be able to recall existing knowledge and organize ideas

Use of appropriate methods:

Justification for choosing the topic using Brainstorming activity:

In home appliances many sensors are used for safety and precise calculation to achieve some task. The students already learned many sensors and their working in previous chapter. So I asked them to prepare and share their ideas.

Effective presentation

Details of Implementation:

The process begins with a well-defined topic. Students are divided into groups. Each participant in a group creates an individual brainstorm around the topic. Each group need to present their views on stage. All the ideas are then merged into one large idea map. During this consolidation phase the participants may discover a common understanding of the issues as they share the meanings behind their ideas. As the sharing takes place, new ideas may arise by the association. Those ideas are added to the map as well. Then ideas are generated on both individual and group levels.

Innovative Teaching Method Execution

Applications- Home appliances - Brain storming



Significance of Results:

Assessment of Effectiveness/Success of the Activity:

- The success of the activity was evaluated by asking the same question in Internal Assessment test III – Around 80% of students answered correctly.
- Slow learners came forward to explain their ideas.

Reflective Critique:

• Benefits:

Brainstorming builds involvement, commitment, loyalty, and enthusiasm. Participating in the sessions stimulates and unlocks people's creative talents. Brainstorming also builds self-esteem because people are being asked for their participation and their ideas.

• Challenges:

Each group allotted for 5 minutes to present their ideas on stage. But few groups take more than the time allotted.

Changes required for future (if required): Nil

Course Outcome	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO5	2	2	-	-	-	-	-	1	1	1	-	-

CO5: The students will be able to implement the DAQ systems with different sensors for real time applications

References:

1. <https://business.tutsplus.com/articles/top-brainstorming-techniques--cms-27181>
2. Sawney A K and Puneet Sawney, "A Course in Mechanical Measurements and Instrumentation and Control", 12th edition, Dhanpat Rai & Co, New Delhi, 2013.